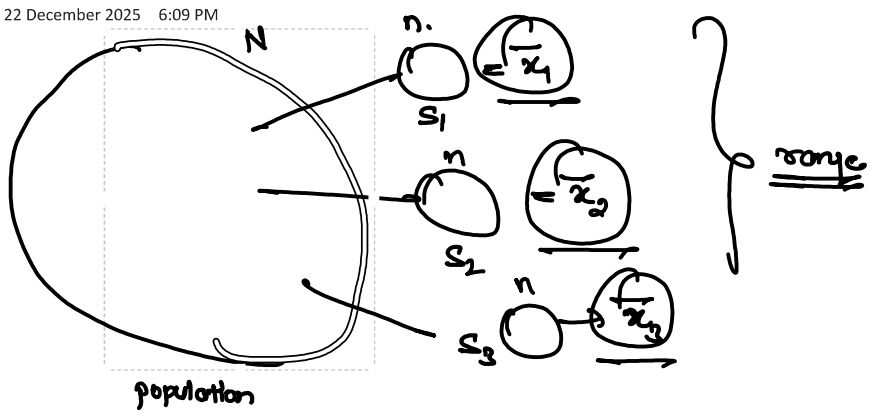
$n \ll N$.

Monday, 22 December 2025 6:09 PM

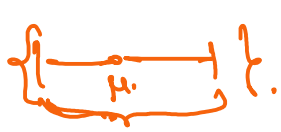


Point estimate → one single value. it does not account for variability.

- Can be misleading.
  - If the sample that you have collected.
    - If it has small sample size.
    - If the collected data is skewed.
    - If there is an outlier in the sample
    - If the data is not a representative sample.
- A point estimate may to too far from the true population parameter.

- No confidence interval.
  - Point estimate → 1 single value ✓
  - Range. → Confidence interval
    - → 90%.
    - → 95%.
    - → 99%.
  - Poor in decision making.

best estimator / best guess + range.



- How to perform estimation / approximation / Inference?

→ Step-1 Take a sample (uniform random sampling.) → sample should be large enough.

→ Step-2 → Calculate the sample statistics.

→ Step-3 Estimate the population parameter using the sample statistics.

There are various estimation techniques.

- Point estimate. → best guess. → no information about uncertainity.
- Confidence Interval estimate. → Range of plausible values.

$$\bar{x} \pm \boxed{x_{\alpha/2} \cdot \frac{s}{\sqrt{n}}}$$

Capture.

- Sampling uncertainity.
- Variability.
- Reliable

- Maximum Likelihood estimator.

Choose parameter value that maximize likelihood of observed data.

n ⊂ N → Total no. of people

n → no. of

- Central limit theorem

It states that

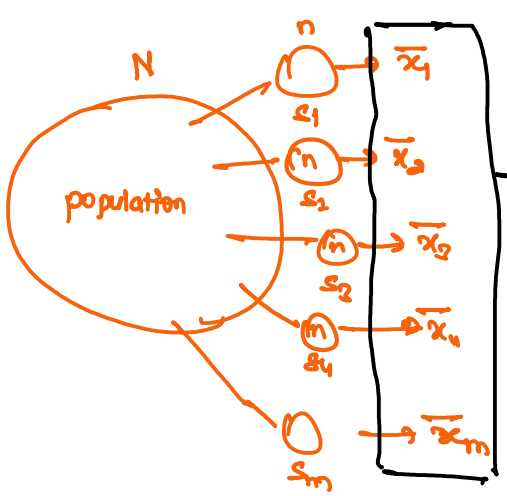
when we take a sufficiently large random sample from any population. (it can follow any distribution), the distribution of the sample mean. becomes approximately normal (bell-shaped), regardless of the population distribution shape.

population → can be anything → skewed, uniform, wierd shaped.
(anything → any type distribution)

→ Take many random samples

→ compute mean of each sample.

→ distribution of these mean → normal.



→ distribution → normal distributed.

- Random sampling.
- Independent observation.
- Sample size large enough

$\boxed{n \geq 30}$

- why it is so important

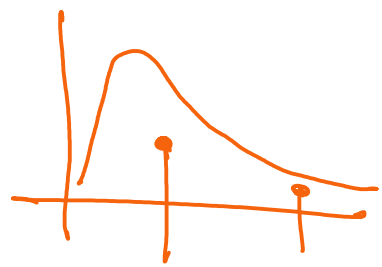
- Because CLT allows us to
  - use normal distribution in real world problems
  - Build confidence interval.
  - Perform hypothesis testing.

Estimator ± (z-value × Standard error)
range

Hypothesis testing.

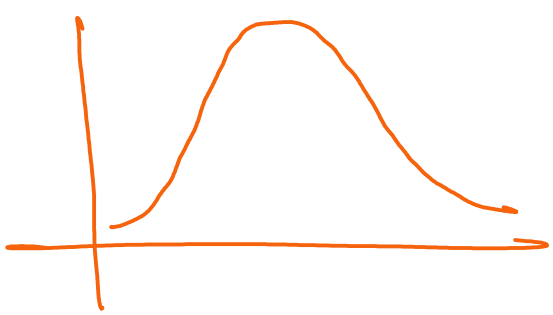
without CLT
↳ we cannot compute
- p-value
- Decide which rare or common.
- Make objective decision.

CLT
↳ a known distribution (normal).
- z-score
  probabilities
- p-values.



CLT turns randomness into reliability.

100 times. { 50 people → calculate one mean
50 }



Extreme (outliers) distort individual values but average smooth them.